

PIYUSH KUMAR MANDAL

Lead Software Engineer

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PROFESSIONAL SUMMARY

Lead Software Engineer with 8.5+ years designing, building and leading delivery of high-throughput, fault-tolerant Java microservices for global banking and insurance clients. Currently technical lead for the API team on Barclays USA's credit-decisioning platform, owning 15–20 Spring Boot services and a team of 8. Track record of scaling production throughput, cutting deployment cycles from hours to minutes, and embedding TDD and observability practice across teams. Hybrid technical lead and people manager — architecture and system design, hands-on delivery, mentoring, agile ceremonies and direct stakeholder ownership. Fluent English; based in Northampton, UK.

TECHNICAL SKILLS

Programming Languages: Java 8 / 17 / 21, J2EE, Python, JavaScript, SQL, HTML, CSS

Frameworks: Spring Boot, Spring Framework, Spring Cloud, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Thymeleaf, Bootstrap

Architecture: Microservices, RESTful Web Services (REST / OAS / Swagger), Event-Driven Architecture, Distributed Systems, System Design at Scale, High Availability, Fault Tolerance, Auto-Scaling

Messaging & Streaming: Apache Kafka, RabbitMQ

Cloud & Containers: Google Cloud Platform (GCP), AWS, Red Hat OpenShift v4, Kubernetes, Docker, App Engine, Cloud SQL

Databases: Oracle, MySQL, Couchbase, MongoDB

DevOps & CI/CD: GitLab CI/CD, Jenkins, GitLab Duo, Git, GitHub, Bitbucket, XLR, Bash / Python automation

Testing & Quality: JUnit, Mockito, Mutation Testing, Test-Driven Development (TDD), Integration & Contract Testing, SonarQube, Code Review

Observability: ELK Stack, AppDynamics, Splunk, Grafana, EsaaS, SLO-based Alerting

Security: OAuth 2.0, JWT, OWASP Top 10, Penetration-Test Remediation

Leadership & Delivery: Agile / Scrum, Sprint Planning, Backlog Grooming, Technical Estimation, Architecture & Design Reviews, Mentoring, On-Call & Incident Management, Stakeholder Management, Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

EXL Service (UK) Limited — Northampton, United Kingdom

Lead Assistant Manager — Lead Software Engineer | Aug 2025 – Present | Client: Barclays USA (GFED – USCB Cards) | Team Size: 8

Barclays issues credit cards on behalf of partner brands; the GFED / USCB Cards platform decides the credit offered to each applicant using bureau data, internal and external fraud signals, and partner-specific decisioning strategies. Transferred to EXL's UK delivery centre to lead the API team.

- Own 15–20 Spring Boot microservices that evaluate card applicants in real time, integrating Experian bureau data, fraud services and partner-specific decisioning strategies.
- Refactored decisioning flows and service boundaries, lifting production throughput from 15K to 25–30K applications per day.
- Lead the API team as technical lead — architecture and requirements analysis, design reviews, on-call rotation, sprint ceremonies and weekly 1:1s.
- Partner directly with Barclays product owners and architects on roadmap planning, technical estimation and stakeholder demos.
- Drive the USCB / Experian cloud migration, including BCP V3 to V4 and ongoing BCP1 to BCP2 platform migrations.

Stack: Java 17/21 · Spring Boot · REST · Microservices · OpenShift / Kubernetes · Docker · Kafka · GitLab CI/CD · Oracle / MySQL

EXL Service — Pune, India

Lead Assistant Manager — Lead Software Engineer | Jul 2024 – Jul 2025 | Client: Barclays USA

- Migrated delivery from Stash and Jenkins to GitLab CI/CD with automated cross-environment promotion — deployment time compressed from 4 hours to 15 minutes.
- Designed Kafka event flows for transaction enrichment, replacing batch processing with sub-second propagation.
- Built the observability layer on ELK, AppDynamics, Grafana and Splunk — structured logs, RED metrics and SLO-based alerting cut mean-time-to-detect by 40% and reduced incident reporting from 1–2 days to under 15 minutes.
- Led the engineering pod through a regulatory compliance delivery, shipped two weeks early with zero post-launch defects.
- Drove TDD adoption across the team — unit-test coverage from 62% to 95%, production defects down 35%, review cycle time roughly halved.

Stack: Java 17 · Spring Boot · Kafka · GitLab CI/CD · OpenShift · ELK · AppDynamics · Grafana · Splunk · JUnit / Mockito

EXL Service — Pune, India

Assistant Manager — Senior Software Engineer | Jun 2022 – Jun 2024 | Client: Barclays USA

- Built and maintained 8+ Spring Boot microservices on OpenShift with full unit, integration and contract test coverage, integrating internal systems and external providers such as Experian.
- Re-platformed a legacy application onto OpenShift V4 — rolling-update strategy, HPA and PDB tuning — delivering 45 TPS at zero downtime and 99.9% uptime, with OAuth introduced during the migration.
- Owned production support rotations — diagnosed and resolved JVM memory leaks and GC pauses under fault-tolerant operation.
- Mentored 2 junior engineers through their first production releases; authored the team's onboarding runbook and PR-review guidelines.
- Partnered with Barclays product owners and architects on quarterly roadmap planning and technical estimation.

Stack: Java 8/17 · Spring Boot · REST · Microservices · OpenShift V4 · Docker · OAuth · Oracle / MySQL · JUnit / Mockito

Cognizant Technology Solutions — Bengaluru, India

Associate Projects | Jul 2021 – Jun 2022 | Client: Google Trust & Safety Devshop

Built extensions and dashboards automating Trust & Safety operations for a 15-person operations team.

- Built browser extensions and dashboards automating Trust & Safety workflows — JavaScript front end, Python back end, Cloud SQL, deployed on Google App Engine.
- Cut recurring overnight manual effort to a 5–10 minute automated run, removing roughly 20 engineering hours per week of toil.
- Owned the full development lifecycle in CWS — requirements, documentation, build, test and deployment across dev, staging and production on GCP.
- Mentored 3 junior engineers on code-quality fundamentals and pair programming; shortened review cycles by around 30%.

Stack: JavaScript · Python · Google Cloud Platform · App Engine · Cloud SQL · REST APIs

ACL Digital (formerly ALTEN Calsoft Labs India Pvt. Ltd.) — India

Senior Software Engineer | Apr 2021 – Jul 2021 | Client: Anglo Gulf Trade Bank (AGTB) | Scrum Team Size: 24

AGTB, an Abu Dhabi commercial bank, was onboarding corporate partners onto its facilities under local regulatory standards; backend team ownership of REST web services.

- Developed, adjusted and tested REST APIs in response to business change, packaging them for deployment across lower environments.
- Designed OAuth 2.0—secured REST APIs for third-party data exposure, closing 100% of API security findings from the prior penetration-test cycle.
- Built a centralised JAR-dependency monitoring dashboard across 12 microservices, surfacing version drift early and trimming production downtime by 15%.

- Ran code reviews for a 5-engineer team and lifted SonarQube maintainability scores from 6.2 to 8.5.

Infosys Limited — Pune / Bhubaneswar, India

Senior Systems Engineer (Engineer III) | Oct 2020 – Apr 2021 | Client: American Express (Third Party Data Gateway) | Scrum Team Size: 12

TPDG received requests from other American Express applications in varied formats and returned responses in the required format after obtaining decisions from the mainframe.

- Refactored the legacy platform into Spring Boot microservices with Kafka event flow, lifting transaction throughput by 35%.
- Implemented OAuth 2.0 and JWT across 15+ REST endpoints, reducing security findings by 20% against the OWASP Top 10.
- Built CI/CD pipelines using Jenkins and reusable XLR templates, compressing release cycles from 2 weeks to 3 days.
- Authored unit, integration and performance test suites — 92% coverage and a ~50% drop in post-deployment defects.
- Gathered requirements with Amex senior engineers and provided final estimates to the Product Owner, hitting 95% of committed delivery dates.

Infosys Limited — Bhubaneswar, India

Systems Engineer (Java Developer) | Nov 2017 – Sep 2020 | Client: The Hartford (Ability Advantage – Self Administration) | Scrum Team Size: 6

Self Administration let corporations onboarded with The Hartford manage all insurance and claim capabilities for their employees from one central system, adjusted in line with legislative change.

- Implemented client-driven feature changes across the Service, DAO and Presentation layers using Spring MVC and Hibernate, cutting batch processing time by 25%.
- Designed and executed comprehensive test plans that closed 50% of known security gaps and validated end-to-end system integrity.
- Diagnosed and resolved P1 production incidents alongside senior engineers, raising overall system reliability by ~15%.
- Profiled JVM and database bottlenecks and shipped fixes that materially improved end-user throughput.

EDUCATION

Maharishi Markandeshwar (Deemed to be University) — Mullana, Ambala, India

Bachelor of Technology (B.Tech.), Computer Science & Engineering | Aug 2013 – May 2017 | CGPA: 8.53

Kendriya Vidyalaya — Bardhaman, West Bengal, India

Senior Secondary (AISSCE), Science | 2013 | 73%

CERTIFICATIONS & COURSES

- **Agentic AI in Action: Hands-On with LangChain** — Skillsoft (Aug 2026)
- **Agentic AI in Action: Tool Use with LangChain** — Skillsoft (Aug 2026)
- **AWS Solutions Architect – Associate** — coursework complete, certification exam in preparation
- **Building Scalable Java Microservices with Spring Boot and Spring Cloud** — Google Cloud / Coursera
- **Google Cloud Fundamentals: Core Infrastructure** — Google Cloud / Coursera
- **Infosys Agile Certified** — Infosys Limited

AWARDS & RECOGNITION

- **Stellar Performer Award – Data Management**, EXL Service (Jul 2026) — recognised for high-impact contribution through ownership, disciplined execution and a solution-oriented approach to complex challenges.
- **Analytics Oscar Award**, EXL Service (Dec 2025).
- **Emerging Talent**, EXL Service (Jan 2023).
- **Insta Award**, Infosys Limited (Jun 2020).